

# VPA ICT Career Pathways Guide

## VPA ICT Career Pathways Guide

Victoria Park Academy | ICT Programme | Career Guidance

A comprehensive guide to careers that build on ICT skills, showing how VPA's curriculum prepares students for each pathway. Includes required skills, qualifications, salary ranges, and VPA preparation steps.

### Career Pathways Overview

English: ICT skills open doors to diverse career paths across four main categories: Digital Literacy careers (using technology effectively), Technical careers (building technology), Creative careers (designing with technology), and Entrepreneurial careers (leading with technology). Each pathway builds on foundations laid in the VPA ICT curriculum.

中文: ICT技能为四大类职业道路打开大门: 数字素养职业(有效使用技术)、技术职业(构建技术)、创意职业(用技术设计)和创业职业(用技术领导)。每条道路都建立在VPA ICT课程奠定的基础之上。

### Pathway 1: Digital Literacy Careers Digital

Administrative & Office Technology

Using productivity and communication tools to manage business operations

| Role                     | Description                                                                   | Key Skills                                                                    | Salary Range (UK/China)                                         |
|--------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Administrative Assistant | Manages schedules, documents, and communications using office software        | Word processing, spreadsheets, email, calendar management, data entry         | <div><div>□ 18k–□ 28k</div><div>□ 8k–□ 15k/month</div></div> /  |
| Office Manager           | Oversees office operations, budgets, and staff coordination                   | Advanced Excel, database management, project management, HR systems           | <div><div>□ 25k–□ 40k</div><div>□ 15k–□ 30k/month</div></div> / |
| Executive Assistant      | Supports senior leadership with complex scheduling, travel, and presentations | Presentation design, travel booking systems, confidentiality, multitasking    | <div><div>□ 30k–□ 50k</div><div>□ 20k–□ 40k/month</div></div> / |
| Virtual Assistant        | Remote administrative support for businesses and entrepreneurs                | Cloud tools, social media management, email marketing, time zone coordination | <div><div>□ 15k–□ 35k</div><div>□ 10k–□ 25k/month</div></div> / |

Required Skills: Microsoft Office / Google Workspace proficiency, touch typing (60+ WPM), email etiquette, basic data analysis, cloud storage management, video conferencing tools

Typical Qualifications: GCSE/IGCSE ICT or equivalent; ECDL (European Computer Driving Licence); NVQ Level 2–3 Business Administration; Bachelor's degree for senior roles

VPA Preparation: Grades 3–8 word processing and spreadsheet units; IGCSE ICT document production and data analysis; touch typing practice in G3–5; presentation skills across all years

Digital Marketing & Communications

Using digital channels to promote products, services, and brands

| Role                       | Description                                                               | Key Skills                                                               | Salary Range (UK/China)        |
|----------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------|--------------------------------|
| Social Media Manager       | Manages brand presence across social platforms, creates content calendars | Content creation, analytics, community management, trend awareness       | £22k-£40k /<br>¥12k-¥25k/month |
| SEO Specialist             | Optimises websites to rank higher in search engine results                | Keyword research, HTML/CSS basics, analytics, content strategy           | £25k-£45k /<br>¥15k-¥30k/month |
| Content Strategist         | Plans and oversees content creation across digital channels               | Writing, data analysis, audience research, CMS management, multimedia    | £30k-£55k /<br>¥20k-¥40k/month |
| Email Marketing Specialist | Designs and executes email campaigns, analyses open rates and conversions | Copywriting, A/B testing, CRM tools, automation platforms, data analysis | £24k-£42k /<br>¥14k-¥28k/month |

**Required Skills:** Social media platforms, analytics tools (Google Analytics), content management systems, basic HTML/CSS, copywriting, image editing, video editing, data interpretation

**Typical Qualifications:** Bachelor's in Marketing, Communications, or Business; Google Analytics Certification; HubSpot Content Marketing Certification; Meta Blueprint Certification

**VPA Preparation:** Multimedia units (G6-8); web design (G6-8); digital citizenship and online safety (all years); data analysis and chart creation (G6-10); presentation and communication skills

## Pathway 2: Technical Careers Technical

### Web Development

Building and maintaining websites and web applications

| Role                 | Description                                              | Key Skills                                                                   | Salary Range (UK/China)        |
|----------------------|----------------------------------------------------------|------------------------------------------------------------------------------|--------------------------------|
| Front-End Developer  | Builds user interfaces using HTML, CSS, and JavaScript   | HTML5, CSS3, JavaScript, React/Vue/Angular, responsive design, accessibility | £28k-£55k /<br>¥18k-¥40k/month |
| Back-End Developer   | Builds server-side logic, databases, and APIs            | Python, Node.js, Java, SQL, REST APIs, cloud services, security              | £32k-£65k /<br>¥22k-¥50k/month |
| Full-Stack Developer | Works across both front-end and back-end                 | All front-end and back-end skills, DevOps basics, deployment, testing        | £35k-£75k /<br>¥25k-¥60k/month |
| Web Designer         | Designs visual layouts and user experiences for websites | UI/UX design, Figma/Adobe XD, HTML/CSS basics, typography, colour theory     | £25k-£50k /<br>¥15k-¥35k/month |

**Required Skills:** HTML, CSS, JavaScript, at least one back-end language, version control (Git), database design, responsive design, web accessibility, performance optimisation

**Typical Qualifications:** Bachelor's in Computer Science or Software Engineering; bootcamp certificates (Le Wagon, General Assembly); portfolio of projects; GitHub profile

**VPA Preparation:** HTML/CSS web design (G6-8); IGCSE ICT web authoring (G9-10); JavaScript introduction (G8); database design (G8-10); programming fundamentals in Python (G6-10)

### Data Analysis & Data Science

Extracting insights from data to support decision-making

| Role                          | Description                                                        | Key Skills                                                            | Salary Range (UK/China)        |
|-------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------------|--------------------------------|
| Data Analyst                  | Cleans, analyses, and visualises data to answer business questions | Excel, SQL, Python/R, Tableau/Power BI, statistics, data cleaning     | £28k–£50k /<br>¥18k–¥35k/month |
| Business Intelligence Analyst | Builds dashboards and reports for business stakeholders            | SQL, ETL tools, data warehousing, Tableau/Power BI, business acumen   | £32k–£60k /<br>¥22k–¥45k/month |
| Data Scientist                | Builds predictive models and machine learning algorithms           | Python/R, machine learning, statistics, deep learning, big data tools | £40k–£90k /<br>¥30k–¥80k/month |
| Database Administrator        | Manages, maintains, and secures organisational databases           | SQL, database design, backup/recovery, performance tuning, security   | £35k–£65k /<br>¥25k–¥50k/month |

**Required Skills:** SQL, Python or R, statistics, data visualisation, Excel, critical thinking, domain knowledge, communication (explaining data to non-technical audiences)

**Typical Qualifications:** Bachelor's in Mathematics, Statistics, Computer Science, or Economics; Master's for Data Science roles; Google Data Analytics Certificate; IBM Data Science Professional Certificate

**VPA Preparation:** Spreadsheet modelling and data analysis (G6–10); database design and SQL (G8–10); IGCSE ICT data analysis unit; statistics in Mathematics; Python programming (G6–10)

#### Software Engineering & Development

Designing, building, and maintaining software applications and systems

| Role                 | Description                                                 | Key Skills                                                               | Salary Range (UK/China)        |
|----------------------|-------------------------------------------------------------|--------------------------------------------------------------------------|--------------------------------|
| Software Developer   | Writes code for applications across platforms               | Python, Java, C#, JavaScript, OOP, testing, version control, APIs        | £30k–£60k /<br>¥20k–¥45k/month |
| Mobile App Developer | Builds applications for iOS and Android                     | Swift (iOS), Kotlin/Java (Android), React Native, Flutter, UX principles | £32k–£65k /<br>¥22k–¥50k/month |
| DevOps Engineer      | Bridges development and operations, manages CI/CD pipelines | Linux, Docker, Kubernetes, AWS/Azure, scripting, automation, monitoring  | £40k–£80k /<br>¥30k–¥70k/month |
| QA Engineer          | Ensures software quality through testing and automation     | Manual testing, automated testing (Selenium), bug tracking, scripting    | £28k–£55k /<br>¥18k–¥40k/month |

**Required Skills:** 1–2 programming languages, data structures, algorithms, OOP, version control, testing, debugging, system design, cloud basics

**Typical Qualifications:** Bachelor's in Computer Science or Software Engineering; coding bootcamps; industry certifications (AWS, Azure); strong GitHub portfolio

**VPA Preparation:** IB DP Computer Science (G11–12); Python programming (G6–10); algorithm design and pseudocode (G6–10); problem-solving in Mathematics; IGCSE ICT programming logic

## Pathway 3: Creative Careers Creative

#### Graphic Design & Visual Communication

Creating visual content for print, digital, and multimedia platforms

| Role                     | Description                                                             | Key Skills                                                                  | Salary Range (UK/China)        |
|--------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------|--------------------------------|
| Graphic Designer         | Creates visual concepts for branding, marketing, and publications       | Adobe Creative Suite, typography, colour theory, layout, branding           | £22k-£45k /<br>¥12k-¥30k/month |
| Brand Designer           | Develops visual identity systems for organisations                      | Logo design, brand guidelines, market research, storytelling, strategy      | £30k-£60k /<br>¥20k-¥45k/month |
| Motion Graphics Designer | Creates animated graphics for video, film, and digital media            | After Effects, Cinema 4D, storytelling, timing, sound design basics         | £28k-£55k /<br>¥18k-¥40k/month |
| Illustrator              | Creates original artwork for books, magazines, advertising, and digital | Drawing, digital illustration (Procreate, Illustrator), visual storytelling | £20k-£50k /<br>¥10k-¥35k/month |

**Required Skills:** Adobe Photoshop, Illustrator, InDesign; drawing and illustration; colour theory; typography; composition; client communication; time management

**Typical Qualifications:** Bachelor's in Graphic Design, Visual Communication, or Fine Arts; strong portfolio; Adobe Certified Professional; D&AD New Blood entries

**VPA Preparation:** Digital art and image editing (G3-8); multimedia production (G6-10); Art and Design GCSE/IGCSE; photography club; design thinking projects

**User Experience (UX) & User Interface (UI) Design**

Designing intuitive, accessible, and enjoyable digital experiences

| Role             | Description                                                  | Key Skills                                                                     | Salary Range (UK/China)        |
|------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------|
| UX Designer      | Researches user needs and designs product experiences        | User research, wireframing, prototyping, usability testing, Figma, psychology  | £30k-£60k /<br>¥20k-¥45k/month |
| UI Designer      | Designs visual interfaces for apps and websites              | Visual design, design systems, Figma/Sketch, interaction design, animation     | £28k-£55k /<br>¥18k-¥40k/month |
| UX Researcher    | Conducts research to understand user behaviours and needs    | Qualitative/quantitative research, interviews, surveys, data analysis, empathy | £32k-£65k /<br>¥22k-¥50k/month |
| Product Designer | End-to-end design of digital products from concept to launch | UX + UI + business strategy, prototyping, A/B testing, collaboration           | £35k-£75k /<br>¥25k-¥60k/month |

**Required Skills:** Figma, Sketch, Adobe XD; user research methods; wireframing and prototyping; usability testing; information architecture; accessibility (WCAG); HTML/CSS basics; empathy and communication

**Typical Qualifications:** Bachelor's in Design, HCI, Psychology, or Computer Science; UX bootcamps (General Assembly, CareerFoundry); portfolio with case studies; Nielsen Norman Group courses

**VPA Preparation:** Web design (G6-8); digital citizenship and empathy (all years); Psychology GCSE/IB; design thinking in Project-Based Learning; HTML/CSS for understanding developer constraints

**Video Production & Digital Media**

Creating video content for entertainment, education, and marketing

| Role                       | Description                                                       | Key Skills                                                                   | Salary Range (UK/China)        |
|----------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------------|--------------------------------|
| Video Editor               | Edits raw footage into polished videos for various platforms      | Premiere Pro, DaVinci Resolve, storytelling, pacing, colour grading, audio   | £24k–£50k /<br>¥15k–¥35k/month |
| Content Creator / YouTuber | Produces and publishes video content for online audiences         | Filming, editing, scripting, audience engagement, analytics, monetisation    | Variable /<br>¥10k–¥50k+/month |
| Broadcast Engineer         | Manages technical infrastructure for live and recorded broadcasts | Video systems, streaming protocols, signal processing, troubleshooting       | £30k–£55k /<br>¥20k–¥40k/month |
| Animator                   | Creates animated content for film, TV, games, and web             | 2D/3D animation, After Effects, Blender/Maya, storytelling, character design | £25k–£55k /<br>¥15k–¥40k/month |

**Required Skills:** Video editing software, camera operation, audio editing, storytelling, colour correction, encoding and compression, platform knowledge (YouTube, TikTok, Instagram)

**Typical Qualifications:** Bachelor's in Film, Media Production, or Animation; Adobe Certified Professional; showreel/portfolio; industry internships

**VPA Preparation:** Multimedia and video editing (G6–8); IGCSE ICT multimedia unit; Drama and Media Studies; school news channel; film club; storyboarding in Art

## Pathway 4: Entrepreneurial Careers Entrepreneurial

### Technology Entrepreneurship

Starting and leading technology-driven businesses

| Role                    | Description                                                | Key Skills                                                                                          | Salary Range (UK/China)                    |
|-------------------------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------|
| Tech Startup Founder    | Creates and leads a technology startup from idea to scale  | Vision, fundraising, product management, leadership, resilience, coding (optional but valuable)     | Variable (equity-based) /<br>¥0–¥1M+/month |
| Product Manager         | Defines product vision and coordinates teams to deliver it | Market research, user stories, Agile/Scrum, data analysis, communication, prioritisation            | £35k–£80k /<br>¥25k–¥70k/month             |
| E-commerce Entrepreneur | Builds and operates online retail businesses               | Platform management (Shopify, Amazon), digital marketing, supply chain, customer service, analytics | Variable /<br>¥15k–¥100k+/month            |
| EdTech Developer        | Creates educational technology products and platforms      | Pedagogy, UX design, programming, content creation, assessment design, accessibility                | £30k–£65k /<br>¥20k–¥50k/month             |

**Required Skills:** Business acumen, financial literacy, leadership, communication, problem-solving, risk assessment, digital marketing, basic coding (advantageous), networking, resilience

**Typical Qualifications:** Bachelor's in Business, Engineering, or Computer Science; MBA (for senior roles); incubator/accelerator programmes (Y Combinator, Techstars); proven track record

VPA Preparation: Business Studies GCSE/IB; Enterprise Club; design thinking and project-based learning; public speaking and debate; ICT skills for building MVPs; Mathematics for financial modelling

## Salary Comparison by Region

| Role              | UK (London) | UK (National) | China (Shanghai/Beijing) | China (Tier 2) |
|-------------------|-------------|---------------|--------------------------|----------------|
| Web Developer     | £35k–£70k   | £28k–£55k     | ¥25k–¥60k/mo             | ¥15k–¥35k/mo   |
| Data Analyst      | £35k–£65k   | £28k–£50k     | ¥22k–¥50k/mo             | ¥12k–¥28k/mo   |
| UX Designer       | £38k–£70k   | £30k–£60k     | ¥25k–¥55k/mo             | ¥15k–¥30k/mo   |
| Graphic Designer  | £28k–£55k   | £22k–£45k     | ¥15k–¥35k/mo             | ¥8k–¥20k/mo    |
| Digital Marketer  | £28k–£55k   | £22k–£42k     | ¥15k–¥35k/mo             | ¥10k–¥22k/mo   |
| Software Engineer | £40k–£90k   | £30k–£65k     | ¥30k–¥80k/mo             | ¥18k–¥45k/mo   |
| Product Manager   | £45k–£95k   | £35k–£80k     | ¥30k–¥80k/mo             | ¥20k–¥50k/mo   |

## VPA ICT → Career Readiness Checklist

| By End of...     | Students Should Be Able To...                                                                                                      | Career Readiness                                                                  |
|------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Grade 5          | Touch type at 30 WPM; create formatted documents and presentations; search safely; explain online safety rules                     | Ready for digital literacy roles with supervision                                 |
| Grade 8          | Build a simple website; write Python programs with loops and conditionals; design a database; edit images and video                | Ready for creative and technical internships; prepared for IGCSE                  |
| Grade 10 (IGCSE) | Produce examination-standard documents; build data models; write SQL; create multi-page websites; understand networks and security | Ready for entry-level technical roles; fully prepared for IB DP CS                |
| Grade 12 (IB DP) | Design and implement OOP solutions; analyse algorithms; understand systems architecture; complete a 2000-word CS investigation     | Ready for university CS programmes; competitive for internships at tech companies |

## Further Education Pathways

UK: University degrees in Computer Science, Software Engineering, Digital Media, or Business Information Systems. Apprenticeships (Degree Apprenticeships) combine work and study. Notable: Imperial College London, University of Cambridge, University of Oxford, University of Manchester.

US: Bachelor's in CS, Information Science, or HCI. Strong emphasis on portfolio and internships. Notable: MIT, Stanford, Carnegie Mellon, UC Berkeley, Georgia Tech.

China: Bachelor's in 计算机科学与技术 (Computer Science), 软件工程 (Software Engineering), or 数字媒体技术 (Digital Media Technology). Notable: Tsinghua University, Peking University, Shanghai Jiao Tong University, Zhejiang University.

Hong Kong: Bachelor's in CS, Information Engineering, or Multimedia Computing. Notable: University of Hong Kong (HKU), Hong Kong University of Science and Technology (HKUST), Chinese University of Hong Kong (CUHK).

深造途径：英国——计算机科学、软件工程、数字媒体或商业信息系统学位；美国——CS、信息科学或HCI学士学位，重视作品集和实习；中国——计算机科学与技术、软件工程或数字媒体技术学士学位；香港——计算机科学、信息工程或多媒体计算学士学位。

□ Victoria Park Academy | ICT Programme | Career Pathways Guide v1.0Salary data indicative, based on 2024 market research. Actual salaries vary by experience, location, and company.